

Responses to review comments:

Reviewer: 1

Comments to the Author

The manuscript by Lin et al. introduces Graph-KIR, a novel graph-based computational tool for KIR gene copy number estimation and high-resolution allele calling from short-read WGS data. The tool's strengths are its innovative per-sample graph-based methodology and strong performance in gene-level identification. However, claims of high-resolution allele calling accuracy are not sufficiently benchmarked against the current state-of-the-art. Furthermore, the evaluation on real-world data relies on filtering that limits claims of practical utility. The work requires major revisions to substantiate its claimed advantages before it can be recommended for publication.

--- MAJOR COMMENTS ---

1. The manuscript requires a direct benchmark against the state-of-the-art Geny tool.

- I recognize the Graph-KIR preprint was originally posted in 2023, before the publication of Geny (Zhou et al., 2024). However, Geny now represents the benchmark for accuracy, particularly at the functionally significant 3-digit resolution:

- * Geny: 90.9% precision, 95.3% recall (40 HPRC samples)
- * Graph-KIR: 76.4% accuracy (44 HPRC samples)

- The Geny authors stated they tried Graph-KIR but "were unable to get satisfactory results". While this claim is weak as the Geny authors did not provide additional evidence, it does need to be taken into account. Due to the published accuracy values of Geny, I cannot support the authors claims of that their tool is more accurate than the competition (eg "the results showcase that Graph-KIR remains superior to the alternative regarding of accuracy at all resolutions while maintaining a low number of false negatives"). Ultimately, the lack of reciprocal benchmarking by either group creates a significant gap in comparative evaluation and needs to be addressed.

Response:

We thank the reviewer for this essential comment. We acknowledge that Geny (Zhou et al., 2024) has established a high standard for KIR typing. In this revision, we have performed a comprehensive, reciprocal benchmark against Geny using both simulated data and the full HPRC dataset.

To address the discrepancies noted by the reviewer, we have implemented the following improvements and clarifications:

- (1) **Standardization of Metrics:** We have corrected the terminology in our manuscript. The metric previously labeled as "Accuracy" was, in fact, Recall. We now provide a full suite of metrics—Recall, Precision, and F1-score—for all tools to ensure a transparent comparison. Notably, our analysis of the Geny source code revealed that its internal recall calculation excludes "mis-paired" alleles from the denominator, which can lead to an overestimation of performance. To ensure a rigorous benchmark, we have recalculated Geny's performance using the standard definitions above for all comparisons in this study.
- (2) **Benchmark on HPRC Data (44 Samples):** We conducted the evaluation based on the 44 HPRC samples, larger than the 40 samples used in the Geny publication. The results on the simulated data (Tables 1 and 3) and the real data (Tables 2 and 4) demonstrate a nuanced performance landscape:
 - A. **Copy Number Estimation:** Graph-KIR consistently outperforms existing methods, providing the robust foundation necessary for accurate allele calling. (Table 1 and Table 2)
 - B. **7-digit Resolution:** Graph-KIR remains superior to other tools, including Geny, in resolving non-coding variations. (Table 3 and Table 4)
 - C. **Functional Typing (3-digit):** Graph-KIR achieves higher Recall, whereas Geny demonstrates higher Precision. (Table 3 and Table 4)
- (3) To address the "unsatisfactory results" mentioned by the Geny authors, we identified that a primary source of confusion was the inclusion of non-coding variants in functional (3-digit) calls. In this revision, we emphasize Graph-KIR's 'exon-only' configuration. This option restricts the variant set to exonic regions, ensuring that our 3-digit calls align strictly with the functional nomenclature used by Geny and other tools.
- (4) We have tempered our language to reflect these findings. Rather than claiming absolute superiority in every metric, we now specify that Graph-KIR provides the highest resolution (7-digit) accuracy and the most robust copy number estimation currently available, while remaining competitive at functional resolutions.

2. Performance evaluation on HPRC data overstates the tool's practical utility due to the filtering of complex alleles.

- The authors report their highest accuracy only after filtering out novel alleles, fusion events, or other complex variations. A user analyzing a new sample will not know a priori if such complexities exist. Reporting accuracy on a "cleaned" dataset does not reflect the tool's performance in a true discovery setting. The authors should present results on the complete, unfiltered HPRC dataset and discuss the tool's performance and limitations with

these challenging features, providing specific examples of what happens in these cases.

Response:

We thank the reviewer for this critical perspective on "discovery setting" performance. We agree that reporting only filtered results does not fully represent the tool's utility in a real-world clinical or research context where the underlying genomic complexity is unknown.

In the revised manuscript, we have made the following changes:

- (1) Unfiltered Benchmarking: We now present the performance of Graph-KIR, PING, and Geny on the complete, unfiltered HPRC dataset (44 samples) in [Table 4](#). This provides a baseline for the tool's performance in a true discovery setting, including all novel alleles and complex fusion events.
- (2) We have repositioned the "filtered" results as a secondary analysis ([Supplementary Table S2](#)). This serves as a diagnostic evaluation to quantify exactly how much specific features, such as gene fusions or novel polymorphisms, impact calling accuracy for each tool.
- (3) We emphasize that even in the presence of novel variations, Graph-KIR's graph-based alignment maintains higher 7-digit resolution accuracy than alternative methods. We anticipate that this advantage will become increasingly critical as the IPD-KIR database expands to incorporate a broader catalog of novel alleles in the near future.

3. The simulation methodology may not adequately capture the complexity of the KIR locus.

- The current simulation generates reads directly from KIR allele sequences, excluding inter-gene sequences. This simplifies the read mapping challenge by removing homologous sequences and segmental duplications. The large discrepancy in simulated vs real data accuracy further supports the notion that the simulations are not sufficiently representative (91% vs 77.8%). A more robust validation would simulate reads from the entire ~150kb KIR genomic locus, and swap out alleles from the database within the gene locations.

Response:

We appreciate the reviewer's constructive critique regarding our simulation strategy. We acknowledge that simulating reads directly from cDNA or isolated gene sequences can underestimate the challenges posed by the high homology and repetitive nature of the intergenic regions within the ~150kb KIR cluster. Nevertheless, these simulations still demonstrate clear performance differences between the various methods ([Tables 1 and Table 3](#)).

To test whether intergenic reads contribute to the discrepancy between simulated (91.79% F1-score) and real-world (68.67% F1-score) performance, we conducted an experiment where we filtered out reads mapping to intergenic regions (based on GRCh38

coordinates) in the 44 HPRC samples. We found that this filtering process had no significant impact on the allele-calling performance (data not shown). This suggests that while intergenic homology is a known challenge for mapping, the performance "gap" is more likely driven by other complex structural variations and novel polymorphisms present in the HPRC assemblies rather than intergenic noise alone.

We have now explicitly detailed these "failure modes" in [Supplementary Table S2](#). Both Graph-KIR and Geny are similarly impacted by features such as: gene fusion events, large segmental duplications, or novel alleles.

We also considered using the simulation framework proposed by Zhou et al. (Geny). However, we observed that their framework generates and evaluates reads for each haploid assembly separately. In contrast, real-world WGS data is derived from diploid samples, where reads from two different haplotypes must be de-convoluted simultaneously. We believe our simulation, while simplified in terms of intergenic sequence, more accurately captures the diploid de-convolution challenge, which is a primary hurdle in KIR typing.

4. The ground truth evaluation methodology is unclear

- The authors state that the HPRC ground truth evaluation was taken from Hung et. al. 2024 "Genetic complexity of killer-cell immunoglobulin-like receptor genes in human pangenome assemblies" - it is not immediately clear that this is the SKIRT paper. The authors should state clearly whether SKIRT was used, as is available on github, for the ground truth evaluation.

- The methods for HPRC ground truth calling in the Graph-KIR github do not include the `kelvin\_kir\_v1\_0.csv` file that contains the HPRC ground truth allele calls necessary to reproduce the results. I could not find that file in any of the supplemental code, files or zenodo deposits from Hung et. al. The authors should ensure their ground truth methodology is reproducible, either by including all files or directly users how to use SKIRT to reproduce them

Response:

We thank the reviewer for pointing out the need for greater clarity regarding our ground truth data. We have made the following updates to the manuscript and our public repository to ensure full transparency and reproducibility:

1. We have revised the [section 2.8](#) to state clearly that the HPRC ground truth annotations, the Supplemental Table S2 of Hung et al., were generated using SKIRT (Hung et al., 2024).
2. We apologize for the missing file in our previous repository. We have now uploaded the comprehensive ground truth file, [hprc_summary_v1_2_e.tsv](#), to the Graph-KIR GitHub repository. This file contains the finalized HPRC ground truth allele calls

utilized in our evaluation, corresponding to [Supplementary Table S2 from Hung et al. \(2024\)](#). Furthermore, we have provided the scripts for calculating Recall, Precision, and F1-score across all resolutions within our GitHub repository.

3. We selected SKIRT as our ground truth framework based on its rigorous adherence to IPD-KIR nomenclature and its high performance relative to other tools:
 - A. SKIRT utilizes a sequence-based hierarchical classification. The first three digits are defined by protein sequence identity, digits four and five distinguish synonymous variants within the CDS, and the final two digits characterize non-coding variations.
 - B. In our validation of the 47 HPRC genome assemblies, SKIRT demonstrated: 99.7% concordance with Immuannot, while successfully identifying three gene fusion events that Immuannot failed to detect. BAKIR showed significantly lower concordance with both SKIRT (94.2%) and Immuannot (93.9%) and was only evaluated on 26 of the 47 HPRC samples.
4. By providing the raw annotation files and clearly defining the hierarchical logic of the SKIRT pipeline, we ensure that our evaluation methodology is both technically sound and fully reproducible by the community.

--- MINOR COMMENTS ---

5. The github README states the current version is tagged with "v2.0" however this tag does not exist in the repo.

Response:

We have officially tagged the current stable release as v2.0 within the repository. This allows users to easily clone or download specific stable versions using standard Git commands.

Responses to review comments:

Reviewer: 2

Comments to the Author

The manuscript introduces Graph-KIR, a graph-based method for KIR copy number estimation and allele typing from short-read sequencing data. The KIR locus is one of the most challenging regions of the human genome due to its high polymorphism, structural variation, and paralogous gene families, and the development of tools tailored to this complexity is both timely and important. The paper is generally well-written and provides a clear description of the pipeline, as well as comparisons to existing approaches. However, I have some concerns outlined below:

Major:

- Merging highly similar genes (KIR2DL1/2DS1, KIR2DL5A/5B) reduces false discovery but risks masking biologically relevant distinctions.

Response:

By utilizing a merged MSA for paralogous pairs like KIR2DL5A/B, we allow the graph-based model to evaluate all potential variations across both loci simultaneously. This prevents the loss of reads that might otherwise be discarded due to mapping ambiguity between nearly identical references.

Following the alignment phase, Graph-KIR performs allele calling for each gene independently. The tool utilizes locus-specific diagnostic variations within the graph to distinguish between the genes. Consequently, the final output clearly reports separate allele types for KIR2DL5A and KIR2DL5B (or KIR2DL1 and KIR2DS1).

This approach ensures that we reduce false negatives caused by mapping interference while fully preserving the biologically relevant distinctions between separate loci.

- The model assumes uniform sequencing depth and discards multi-mapped reads, which could bias CN estimates in low-complexity regions.

Response:

We thank the reviewer for this technical critique. We would like to clarify our strategy for handling read depth and multi-mapping:

- (1) We intentionally discard multi-mapped reads because their inclusion often leads to a significant over-estimation of local depth, introducing unpredictable noise that biases copy number (CN) estimation. By focusing on uniquely assigned reads within the graph, we ensure a more reliable baseline.
- (2) We acknowledge that removing these reads may lead to a slight under-estimation of depth for certain high-homology genes. To address this, Graph-KIR does not rely

on a fixed depth value. Instead, we model the depth variation using a normal distribution based on estimated mean and standard deviation. This distribution-based approach provides the necessary tolerance for potential under-estimation, allowing for robust CN calling.

- (3) Our results across diverse datasets confirm that this statistical tolerance effectively compensates for the removal of multi-mapped reads, as evidenced by our high accuracy in CN estimation compared to ground truth.

- The authors use hg19 (hs37d5) instead of hg38, citing complications with hg38 scaffolds. This choice is outdated and may limit adoption in future studies... Most recent studies and reference resources have moved to hg38 or T2T-CHM13; reliance on hg19 may hinder reproducibility and comparability.

Response:

We thank the reviewer for this important observation regarding the reference genome version. We acknowledge that hg38 and T2T-CHM13 are the current standards for most genomic analyses. Our initial use of hg19 (hs37d5) was a deliberate choice due to the extreme complexity of the KIR region; hg38 contains numerous KIR-related alternative scaffolds that require cautious handling to avoid multi-mapping artifacts and biased read depth.

However, we would like to clarify that Graph-KIR is fully compatible with hg38. To address the reviewer's concern, we performed additional testing using a scaffold-free version of hg38 as the reference for retrieving KIR-related reads. Our results demonstrate:

- (a) The performance of Graph-KIR when using hg38 is consistent with the results obtained using hg19. This indicates that the underlying graph-based approach is robust to the reference version used for initial read recruitment.
- (b) The users of Graph-KIR can specify the reference version (hg19 or hg38) used. Graph-KIR then adjusts its retrieval coordinates accordingly.

We have added a note in the [section 2.3](#) of the revised manuscript to explicitly state that both hg19 and hg38 are supported and yield comparable performance.

- Simulated data are fixed at 30x, but real samples vary. A systematic downsampling experiment (e.g., 5x–30x) would be important to establish the tool's performance at varying depths.

Response:

We thank the reviewer for highlighting the importance of depth-dependency analysis. While systematic downsampling of simulated data can provide a theoretical baseline, we believe that evaluating performance across real-world sequencing depth variation provides a more rigorous and clinically relevant assessment of the tool's robustness.

To address this, we utilized the inherent depth variability in the 44 HPRC samples, which ranges from 17x to 70x (as detailed in [Supplementary Table S1](#)). We categorized these samples into three cohorts (<30x, 30–40x, and >40x) to evaluate performance across functional and genomic resolutions:

Our results in [Supplementary Table S3](#) indicate that a reduction in sequencing depth to below 30x does not significantly compromise calling accuracy. This suggests that Graph-KIR's graph-based recruitment is highly efficient even with limited data.

Interestingly, we observed slightly lower performance in the highest-depth group (>40x). Our investigation revealed that this group consists primarily of early-released HPRC samples. These samples might be characterized by lower base-calling quality. Similar trend is observed on the results of Geny. Therefore, we consider that sequence quality, rather than depth, is the primary driver of performance variance in these cases.

Minor:

- Although Graph-KIR improves short-read analysis of KIR, short reads are inherently limited in resolving paralogous genes and complex haplotypes compared to long-read or graph-based approaches. The authors should briefly acknowledge this limitation in the discussion.

Response:

We agree that short-read sequencing has physical limitations in resolving the most complex KIR structural variations. We have added a concise acknowledgment of this in the [Discussion section](#), noting that while long-reads offer superior phasing, Graph-KIR remains a critical tool for high-resolution analysis of existing large-scale short-read datasets.

- The sentence “When evaluated with both simulated and real data, the superior performance of Graph-KIR over other existing KIR tools” is incomplete and should be revised for grammatical clarity.

Response:

Thank you for the correction. We fixed the sentence in the revised manuscript.

Furthermore, we have updated the manuscript to further emphasize the primary advantages of Graph-KIR. Specifically, we highlight its superior performance in copy number estimation and 7-digit allele calling. While many existing tools are optimized for 3-digit functional typing, Graph-KIR's graph-based approach allows for a more granular characterization of the KIR region, providing high-accuracy results at the full genomic sequence level (7-digit) and robust copy number profiles across both simulated and real datasets.

- “full-resolution (7-digit)” in the abstract could confuse readers unfamiliar with KIR/HLA

nomenclature. A short clarifier would help.

Response:

We appreciate the reviewer's suggestion to clarify the nomenclature for a broader audience. We recognize that while "7-digit" is a standard term within the KIR and HLA communities, it may be ambiguous to readers in other fields. In the revised abstract, we have added a clarifier to explain that 7-digit resolution refers to the genomic sequence level, which includes both coding (exonic) and non-coding (intronic/UTR) variations.

Revised Text (**Abstract**):

"In this study, we introduce Graph-KIR, a tool designed to estimate gene copy numbers and predict full-resolution (7-digit, encompassing both coding and non-coding sequence variations) KIR alleles from a whole-genome sequencing (WGS) sample."